Coding & Solution

Date	7 JULY 2026
Team ID	
Project Name	Human Development Index (HDI) Predictor – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/AP24110010600/Human-Development-Index-HDI-Predictor-A-Comprehensive-Measure-of-Well-Being
Programming Language(s)	Python 3, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Gunicorn
Key Features Implemented	• Predicts Human Development Index (HDI) using Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. •Machine Learning model for HDI prediction. • Interactive web interface. • Prediction history storage and display. • About Project page. • Input validation and error handling. • Public deployment on Railway.

Pending / Incomplete Features	• User authentication. • Data visualization dashboard. • Export prediction results to PDF/CSV. • Automatic dataset updates from UNDP.
Setup / Run Instructions	1. Clone the GitHub repository. 2. Open the project in VS Code. 3. Create a virtual environment. 4. Install dependencies using `pip install -r requirements.txt`. 5. Run the application using `python app.py`. 6. Open http://127.0.0.1:5001 locally or access the deployed Railway URL.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Human Development Index (HDI) Predictor is a machine learning–based web application developed using Flask and Scikit-learn. The application predicts a country's HDI based on key socio-economic indicators and is successfully deployed on Railway for public access. The project follows a modular architecture, uses version control through GitHub, and provides a user-friendly interface for prediction and result management.

Pending / Incomplete Features:

The current version of the Human Development Index (HDI) Predictor successfully implements the core prediction functionality. However, a few advanced features are planned for future enhancement. These include user authentication for secure access, an interactive data visualization dashboard to display HDI trends and analytics, the ability to export prediction results in PDF or CSV format, and automatic synchronization with the latest UNDP datasets to keep the prediction model updated with current data. These enhancements will improve the usability, security, and overall functionality of the application in future releases.